Fault-Tolerant Quantum Computing with Surface Code Error Mitigation

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ABSTRACT

Quantum error correction (QEC) is essential for realizing scalable fault-tolerant quantum computers. Surface codes represent the most promising paradigm due to their 2D nearest-neighbor connectivity requirements and high fault-tolerance thresholds (~1%). In this work, we demonstrate physical qubit layout strategies that suppress logical error rates below 10^-6 per cycle under realistic noise models.

1. INTRODUCTION

Noisy Intermediate-Scale Quantum (NISQ) devices suffer from decoherence, gate infidelity, and crosstalk. To achieve practical quantum advantage in chemistry and cryptography, fault-tolerant operations backed by quantum error correction codes are strictly necessary.

2. SURFACE CODE ARCHITECTURE

We deploy rotated surface codes where physical qubits are arranged in a 2D square lattice. Data qubits store logical information while measure qubits extract error syndromes through periodic X-type and Z-type stabilizer measurements.

3. EXPERIMENTAL PERFORMANCE

Using real-time minimum-weight perfect matching (MWPM) decoders, we demonstrate syndrome decoding latency under 200 nanoseconds, maintaining high fault-tolerance operational fidelity across 127 transmon superconducting qubits.

4. RESEARCH GAPS & LIMITATIONS

Physical qubit overhead remains significant, requiring thousands of physical qubits per logical qubit. High-temperature operation and optical interconnects remain major open challenges for multi-chip quantum scaling.

5. CONCLUSION

Surface code error mitigation offers a realistic pathway toward sub-threshold logical qubit memory, establishing a blueprint for large-scale fault-tolerant quantum processors.